

Arjuna JEE 2025

Maths

Sequence & Series

DPP: 4

Q1 $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$ to ∞ is equal to

- (A) $\frac{1}{2}$
(B) 1
(C) 2
(D) 3

Q2 If sum of an infinite geometric series is $\frac{4}{3}$ and its 1st term is $\frac{3}{4}$, then its common ratio is

- (A) $\frac{7}{16}$ (B) $\frac{9}{16}$
(C) $\frac{1}{9}$ (D) $\frac{7}{9}$

Q3 Three numbers form an increasing G. P. If the middle term is doubled, then the numbers are in A. P. The common ratio of the G. P. is

- (A) $2 - \sqrt{3}$
(B) $2 + \sqrt{3}$
(C) $\sqrt{3} - 2$
(D) $3 + \sqrt{2}$

Q4 If for $0 < \theta < \frac{\pi}{4}$. Let $x = \sum_{n=0}^{\infty} (\sin \theta)^{2n}$ and $y = \sum_{n=0}^{\infty} (\cos \theta)^{2n}$ then $\sum_{n=0}^{\infty} \sin^{2n} \theta \cdot \cos^{2n} \theta =$

- (A) $\frac{xy}{xy+1}$
(B) $\frac{xy}{xy-1}$
(C) $\frac{xy-1}{xy}$
(D) $\frac{xy+1}{xy}$

Q5 If $S = 1 + a + a^2 + \dots \infty$ where $|a| < 1$ then $a =$

- (A) $\frac{S}{S-1}$
(B) $\frac{S}{1-S}$
(C) $\frac{S-1}{S}$

(D) $\frac{1-S}{S}$

Q6 The value of $y = 0.36^{\log_{0.25} \left(\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots \infty \right)}$ is

- (A) 0.9 (B) 0.8
(C) 0.6 (D) 0.25

Q7 If $S = \sum_{k=1}^{\infty} P^k$ where $|P| < 1$ then $\sum_{k=1}^{\infty} P^{2k} =$

- (A) $\frac{S^2}{S-1}$
(B) $\frac{S^2}{2S+1}$
(C) $\frac{S^2}{2S-1}$
(D) $\frac{S^2}{S+1}$

Q8 $\sum_{n=1}^{\infty} 3\left(\frac{1}{2}\right)^{n-1} =$

- (A) 3
(B) $\frac{1}{2}$
(C) 6
(D) 12

Q9 The product $(32)(32)^{1/6}(32)^{1/36} \dots$ to ∞ is

- (A) 16 (B) 32
(C) 64 (D) 62

Q10 The series

$\frac{2x}{x+3} + \left(\frac{2x}{x+3}\right)^2 + \left(\frac{2x}{x+3}\right)^3 + \dots \infty$ will have a definite sum when

- (A) $-1 < x < 3$
(B) $0 < x < 1$
(C) $x = 0$
(D) $x = -3$

Q11



If the roots of the cubic equation

$ax^3 + bx^2 + cx + d$ are in GP, then

- (A) $c^3a = b^3d$ (B) $ca^3 = bd^3$
(C) $a^3b = c^3d$ (D) $ab^3 = cd^3$

Q12 If sum of series

$S = (\sqrt{2} + 1) + 1 + (\sqrt{2} - 1) + \dots \infty$ is given as $\frac{a+b\sqrt{2}}{c}$, (where $a, b, c \in \mathbb{N}$) then the value of $\frac{a+b}{c}$ is

- (A) -1 (B) $\frac{2}{7}$
(C) $\frac{7}{2}$ (D) 7

Q13 The value of $(0.2)^{\log_{\sqrt{5}}\left(\frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \infty\right)}$ is

Q14 Direction : Let $a_1, a_2, \dots$, be an AP and $b_1, b_2, \dots$, be a GP. The sequence $c_1, c_2, c_3, \dots$, is such that $c_n = a_n + b_n \forall n \in \mathbb{N}$. Suppose $c_1 = 1, c_2 = 4, c_3 = 15$ and $c_4 = 2$. The common ratio of GP is

- (A) -2 (B) -3
(C) 2 (D) 3

Directions (15) Read the following passage and answer the given questions.

Comprehension Type Questions

Direction : Let $a_1, a_2, \dots$, be an AP and $b_1, b_2, \dots$, be a GP. The sequence $c_1, c_2, c_3, \dots$, is such that $c_n = a_n + b_n \forall n \in \mathbb{N}$. Suppose $c_1 = 1, c_2 = 4, c_3 = 15$ and $c_4 = 2$.

Q15 The value of $\sum_{i=1}^{20} a_i$ is

- (A) 480 (B) 770
(C) 960 (D) 1040

Q16 If $a = \sum_{n=\theta}^{\infty} x^n, b = \sum_{n=0}^{\infty} y^n$, $c = \sum_{n=0}^{\infty} (xy)^n$, where $[x], [y] < 1$, then
(A) $abc = a + b + c$ (B) $ab + bc = ac + b$
(C) $ac + bc = ab + c$ (D) $ab + ac = bc + a$

Q17 Find a three-digit number whose consecutive digits form a GP. If we subtract 792 from this

number, we get a number consisting of the same digits written in the reverse order. Now, if we increase the second digit of the required number by 2, then the resulting digits will form an AP.

- (A) 319 (B) 913
(C) 931 (D) 391

Q18 One side of an equilateral triangles is 24 cm. The mid-points of its sides are joined to form another triangles whose mid-points, in turn, are joined to form still another triangle. This process continues, indefinitely. Find the sum of the perimeters of all the triangles.

- (A) 121 (B) 144
(C) 169 (D) 196

Q19 If x, y, z is real and $4x^2 + 9y^2 + 16z^2 - 6xy - 12yz - 8zx = 0$ then x, y, z is in

- (A) AP
(B) GP
(C) HP
(D) None

Q20 If 7th term of HP is $\frac{1}{20}$ and 13th term is $\frac{1}{38}$ then its 4th term is equal to

- (A) 11
(B) $\frac{1}{11}$
(C) 12
(D) $\frac{1}{12}$

Q21 Find the sum of series, $6 + 66 + 666 + \dots$ upto n terms.

- (A) $\frac{6}{81}(10^{n+1} - 9n - 10)$
(B) $\frac{2}{27}(10^n - 9n - 1)$
(C) $\frac{6}{81}(10^{n+1} - 9n + 10)$
(D) $\frac{2}{27}(10^{n+1} - 9n + 1)$

Q22 The fifth term of the H.P., $2, 2\frac{1}{2}, 3\frac{1}{3}, \dots$, will be
(A) $5\frac{1}{5}$



- (B) $3\frac{1}{5}$
- (C) $\frac{1}{10}$
- (D) 10

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Answer Key

Q1 (B)

Q2 (A)

Q3 (B)

Q4 (B)

Q5 (C)

Q6 (C)

Q7 (B)

Q8 (C)

Q9 (C)

Q10 (A)

Q11 (A)

Q12 (C)

Q13 4

Q14 (B)

Q15 (C)

Q16 (C)

Q17 (C)

Q18 (B)

Q19 (C)

Q20 (B)

Q21 (A)

Q22 (D)

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